MS MARCO – Official Direct Download

```
#  Passage Collection (8.8M passages ~ 2.3GB)

!wget
https://msmarco.z22.web.core.windows.net/msmarcoranking/collection.tar
.gz
!tar -xvf collection.tar.gz
```

```
--2026-02-23 12:23:07--
https://msmarco.z22.web.core.windows.net/msmarcoranking/collection.tar
.gz
Resolving msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)... 57.150.146.100
Connecting to msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)|57.150.146.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1035009698 (987M) [application/octet-stream]
Saving to: ‘collection.tar.gz.1’

collection.tar.gz.1 100%[===================>] 987.06M  47.7MB/s    in
28s

2026-02-23 12:23:35 (35.6 MB/s) - ‘collection.tar.gz.1’ saved
[1035009698/1035009698]

collection.tsv
```

```
# Queries (Training Queries ~ 1M)

!wget
https://msmarco.z22.web.core.windows.net/msmarcoranking/queries.tar.gz
!tar -xvf queries.tar.gz
```

```
--2026-02-23 12:24:13--
https://msmarco.z22.web.core.windows.net/msmarcoranking/queries.tar.gz
Resolving msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)... 57.150.146.100
Connecting to msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)|57.150.146.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 18882551 (18M) [application/gzip]
Saving to: ‘queries.tar.gz.2’

queries.tar.gz.2    100%[===================>]  18.01M  36.5MB/s    in
0.5s

2026-02-23 12:24:13 (36.5 MB/s) - ‘queries.tar.gz.2’ saved
[18882551/18882551]

queries.dev.tsv
```

```
queries.eval.tsv
queries.train.tsv
```

```
!wget
https://msmarco.z22.web.core.windows.net/msmarcoranking/qrels.train.ts
v
```

```
--2026-02-23 12:24:14--
https://msmarco.z22.web.core.windows.net/msmarcoranking/qrels.train.ts
v
Resolving msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)... 57.150.146.100
Connecting to msmarco.z22.web.core.windows.net
(msmarco.z22.web.core.windows.net)|57.150.146.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10589532 (10M) [text/tab-separated-values]
Saving to: ‘qrels.train.tsv.1’

qrels.train.tsv.1   100%[===================>]  10.10M  38.7MB/s    in
0.3s

2026-02-23 12:24:14 (38.7 MB/s) - ‘qrels.train.tsv.1’ saved
[10589532/10589532]
```

```
import pandas as pd

# Load first 500k passages only
df = pd.read_csv("collection.tsv", sep="\t", header=None,
nrows=500000)
```

```
import pandas as pd

df = pd.read_csv("collection.tsv", sep="\t", header=None,
nrows=500000)
df.columns = ["pid", "passage"]

print(df.shape)
```

```
(500000, 2)
```

```
# PHASE 1 Mini Corpus Load + Basic Retrieval Test

# STEP 1 — Load Mini Corpus (Safe Version)

import pandas as pd

df = pd.read_csv(
    "collection.tsv",
    sep="\t",
    header=None,
    nrows=200000,
```

```
    encoding="utf-8"
)

df.columns = ["pid", "passage"]

print(df.head())
```

```
   pid                                                passage
0    0  The presence of communication amid scientific ...
1    1  The Manhattan Project and its atomic bomb help...
2    2  Essay on The Manhattan Project - The Manhattan...
3    3  The Manhattan Project was the name for a proje...
4    4  versions of each volume as well as complementa...
```

```
# STEP 2 — Install Embedding Model

!pip install sentence-transformers faiss-cpu
```

```
Requirement already satisfied: sentence-transformers in
/usr/local/lib/python3.12/dist-packages (5.2.3)
Requirement already satisfied: faiss-cpu in
/usr/local/lib/python3.12/dist-packages (1.13.2)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(5.0.0)
Requirement already satisfied: huggingface-hub>=0.20.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.4.1)
Requirement already satisfied: torch>=1.11.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(2.10.0+cu128)
Requirement already satisfied: numpy in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(2.0.2)
Requirement already satisfied: scikit-learn in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.6.1)
Requirement already satisfied: scipy in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(1.16.3)
Requirement already satisfied: typing_extensions>=4.5.0 in
/usr/local/lib/python3.12/dist-packages (from sentence-transformers)
(4.15.0)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-
packages (from sentence-transformers) (4.67.3)
Requirement already satisfied: packaging in
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.0)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (3.24.2)
```

```
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (0.28.1)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (6.0.3)
Requirement already satisfied: shellingham in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (1.5.4)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0-
>sentence-transformers) (0.24.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.1.6)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
```

```
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.11.0->sentence-
transformers) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4-
>torch>=1.11.0->sentence-transformers) (1.3.4)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.22.2)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from
transformers<6.0.0,>=4.41.0->sentence-transformers) (0.7.0)
Requirement already satisfied: joblib>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-
transformers) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn->sentence-
```

```
transformers) (3.6.0)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.20.0->sentence-transformers) (4.12.1)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.20.0->sentence-transformers) (2026.1.4)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub>=0.20.0->sentence-transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-
transformers) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub>=0.20.0->sentence-transformers)
(0.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.11.0->sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0-
>sentence-transformers) (3.0.3)
Requirement already satisfied: typer>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from typer-slim->huggingface-
hub>=0.20.0->sentence-transformers) (0.24.0)
Requirement already satisfied: click>=8.2.1 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->huggingface-hub>=0.20.0->sentence-transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->huggingface-hub>=0.20.0->sentence-transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-
slim->huggingface-hub>=0.20.0->sentence-transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0-
>typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0->sentence-
transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer>=0.24.0->typer-slim->huggingface-hub>=0.20.0-
>sentence-transformers) (0.1.2)

# STEP 3 — Create Embeddings (Small Batch)
```

```python
from sentence_transformers import SentenceTransformer
import numpy as np

model = SentenceTransformer("all-MiniLM-L6-v2")

# Sirf passages ka list
passages = df["passage"].tolist()

# Batch encoding (important for memory)
embeddings = model.encode(passages, batch_size=64,
show_progress_bar=True)

print("Embedding shape:", embeddings.shape)
```

```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

{"model_id":"2692b9b00da64017922440cf6dfc9ac0","version_major":2,"version_minor":0}

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                      | Status     |  |
-------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

{"model_id":"cab9845196d54fc6b1a0f333a0a5606d","version_major":2,"version_minor":0}

```
Embedding shape: (200000, 384)
```

```python
# STEP 4 — Build FAISS Index

import faiss

dimension = embeddings.shape[1]

index = faiss.IndexFlatL2(dimension)
index.add(np.array(embeddings))

print("Total indexed:", index.ntotal)
```

```
Total indexed: 200000
```

```python
# STEP 5 — Test Simple Retrieval
```

```python
query = "What is machine learning?"

query_embedding = model.encode([query])

k = 5
distances, indices = index.search(np.array(query_embedding), k)

for i in indices[0]:
    print("\n----")
    print(df.iloc[i]["passage"])
```

```
----
â€Deep learning is a particular kind of machine learning that achieves great power and flexibility by learning to represent the world as nested hierarchy of concepts, with each concept defined in relation to simpler concepts, and more abstract representations computed in terms of less abstract ones.â€

----
If you have often wondered to yourself about the difference between machine learning and deep learning, read on to get a detailed comparison in simple layman language. I explain each of these terms in detail.

----
In a recent article, we demystified some of the technical jargon that's being thrown around these days like 'artificial intelligence', 'SaaS, 'the cloud', and 'deep learning'. While the techies can debate among themselves the difference between 'machine learning' and 'deep learning', we're going to consider the two terms synonymous and henceforth just talk about 'deep learning'. So just what is 'deep learning'? We wanted to understand more, so we came across this excellent TED talk given by Jeremy Howard which finally explains in layman's terms just what deep learning is. If you have 20 minutes, watch the video now and no need to

----
Deep learning refers to artificial neural networks that are composed of many layers. It's a growing trend in ML due to some favorable results in applications where the target function is very complex and the datasets are large. For example in Hinton et al.

----
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```

```python
# STEP 2 — Retrieval Quality Measurement (RCS)
```

```python
# STEP 2.1 — Cosine Similarity Correct

import numpy as np
import faiss

# Normalize embeddings for cosine similarity
faiss.normalize_L2(embeddings)

dimension = embeddings.shape[1]
index = faiss.IndexFlatIP(dimension)  # Inner product for cosine
index.add(embeddings)

print("Total indexed:", index.ntotal)
```

```
Total indexed: 200000
```

```python
# STEP 2.2 — Query Retrieval with Scores

query = "What is machine learning?"

query_embedding = model.encode([query])
faiss.normalize_L2(query_embedding)

k = 5
scores, indices = index.search(query_embedding, k)

print("Top-k similarity scores:", scores[0])
```

```
Top-k similarity scores: [0.57475275 0.5625971  0.54654205 0.5455879
0.5303432 ]
```

```python
# STEP 2.3 — Retrieval Coverage Score (RCS)

rcs = np.mean(scores[0])
print("Retrieval Coverage Score (RCS):", round(float(rcs), 4))
```

```
Retrieval Coverage Score (RCS): 0.552
```

```python
# STEP 3 — Retrieval Stability Metric

import numpy as np

topk_scores = scores[0]

mean_score = np.mean(topk_scores)
std_score = np.std(topk_scores)
gap = topk_scores[0] - topk_scores[-1]

print("Mean:", round(float(mean_score),4))
print("Std Dev:", round(float(std_score),4))
print("Top1-TopK Gap:", round(float(gap),4))
```

```
Mean: 0.552
Std Dev: 0.0153
Top1-TopK Gap: 0.0444
```

```python
# STEP 4 — Weak Retrieval Detection Logic

def retrieval_risk_assessment(mean, std, gap):
    risk_score = 0

    # Coverage risk
    if mean < 0.50:
        risk_score += 2
    elif mean < 0.60:
        risk_score += 1

    # Instability risk
    if std > 0.05:
        risk_score += 1

    # Over-reliance risk
    if gap > 0.10:
        risk_score += 1

    if risk_score == 0:
        return "Low Retrieval Risk"
    elif risk_score == 1:
        return "Moderate Retrieval Risk"
    else:
        return "High Retrieval Risk"


risk = retrieval_risk_assessment(mean_score, std_score, gap)
print("Retrieval Risk Level:", risk)
```

```
Retrieval Risk Level: Moderate Retrieval Risk
```

```python
# STEP 5 — Retrieval-Aware Generation Gate

# STEP 5.1 — Simple Retrieval-Aware Prompt Builder

def build_prompt(query, retrieved_passages, risk_level):

    context = "\n\n".join(retrieved_passages)

    if risk_level == "Low Retrieval Risk":
        instruction = "Answer confidently using the provided
evidence."

    elif risk_level == "Moderate Retrieval Risk":
        instruction = "Answer carefully using only the provided
evidence. If unsure, state uncertainty."
```

```
    else:
        instruction = "Evidence may be weak. Only answer if strongly
supported. Otherwise say insufficient evidence."

    prompt = f"""
You are a factual assistant.

Query:
{query}

Evidence:
{context}

Instruction:
{instruction}

Answer:
"""
    return prompt

# STEP 5.2 — Prepare Retrieved Passages

retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

prompt = build_prompt(query, retrieved_passages, risk)

print(prompt[:1000])  # just preview
```

```
You are a factual assistant.

Query:
What is machine learning?

Evidence:
â€Deep learning is a particular kind of machine learning that achieves
great power and flexibility by learning to represent the world as
nested hierarchy of concepts, with each concept defined in relation to
simpler concepts, and more abstract representations computed in terms
of less abstract ones.â€

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among themselves the difference between 'machine learning' and 'deep
learning', we're going to consider the two terms synonymous and
```

```
henceforth just talk about 'deep learning'. So just what is 'deep
learn
```

```
# STEP 6 — Evidence Alignment Score (EAS)

# STEP 6.1 — Simple Semantic Alignment Scorer

def evidence_alignment_score(answer, retrieved_passages, model):
    evidence_text = " ".join(retrieved_passages)

    answer_embedding = model.encode([answer])
    evidence_embedding = model.encode([evidence_text])

    faiss.normalize_L2(answer_embedding)
    faiss.normalize_L2(evidence_embedding)

    alignment = float((answer_embedding @
evidence_embedding.T).item())

    return alignment

# STEP 6.2 — Test With Sample Answer

sample_answer = "Machine learning is a type of artificial intelligence
that allows systems to learn patterns from data."

alignment_score = evidence_alignment_score(sample_answer,
retrieved_passages, model)

print("Evidence Alignment Score:", round(alignment_score, 4))
```

```
Evidence Alignment Score: 0.5334
```

```
# STEP 7 — Hallucination Probability Score (HPS)

# STEP 7.1 — Basic Hallucination Gap

hallucination_gap = rcs - alignment_score
print("Hallucination Gap:", round(hallucination_gap, 4))
```

```
Hallucination Gap: 0.0185
```

```
# Hallucination Probability Function

def hallucination_probability(rcs, eas):
    gap = rcs - eas

    if gap < 0:
        return 0.1  # Strong grounding
    elif gap < 0.05:
        return 0.3  # Low hallucination risk
    elif gap < 0.10:
        return 0.6  # Moderate risk
```

```
    else:
        return 0.85  # High risk

# Compute HPS

hps = hallucination_probability(rcs, alignment_score)
print("Hallucination Probability Score (HPS):", hps)
```

```
Hallucination Probability Score (HPS): 0.3
```

```
# STEP 8 — Composite Trust Score (CTS)

# STEP 8.1 — Define Stability Score

stability_score = 1 - std_score
print("Stability Score:", round(stability_score,4))
```

```
Stability Score: 0.9847
```

```
# STEP 8.2 — Define Composite Trust Function

def composite_trust_score(rcs, eas, hps, stability):

    w1 = 0.30  # Retrieval Coverage
    w2 = 0.30  # Evidence Alignment
    w3 = 0.25  # Hallucination Inverse
    w4 = 0.15  # Stability

    cts = (
        w1 * rcs +
        w2 * eas +
        w3 * (1 - hps) +
        w4 * stability
    )

    return round(float(cts), 4)

# STEP 8.3 — Compute CTS

cts = composite_trust_score(rcs, alignment_score, hps,
stability_score)

print("Composite Trust Score (CTS):", cts)
```

```
Composite Trust Score (CTS): 0.6483
```

```
# STEP 9 — Multi-Query Evaluation Mode

# STEP 9.1 — Load Query Sample

# Load query file
queries_df = pd.read_csv(
    "queries.train.tsv",
```

```
    sep="\t",
    header=None,
    names=["qid", "query"],
    nrows=1000  # safe load
)

# Random 50 queries
sample_queries = queries_df.sample(50, random_state=42)
["query"].tolist()

print("Total queries selected:", len(sample_queries))
sample_queries[:5]
```

```
Total queries selected: 50

['what year did political parties emerge',
 'what is a running bond',
 'what transponders does cmrc use',
 'et al abbreviation meaning',
 'phone number amazon customer service usa']
```

```
# STEP 9.2 — Evaluation Loop

results = []

for query in sample_queries:

    # Retrieve
    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))

    # Stability
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    # Get passages
    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    # Simple synthetic answer (for now)

    synthetic_answer = "Quantum gravity explains black holes through
string theory."

    # Alignment
    alignment_score = evidence_alignment_score(
        synthetic_answer,
```

```
        retrieved_passages,
        model
    )

    # HPS
    hps = hallucination_probability(rcs, alignment_score)

    # CTS
    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    results.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "CTS": cts
    })

# STEP 9.3 — Convert to DataFrame

results_df = pd.DataFrame(results)

results_df.head()
```

{"summary":"{\n \"name\": \"results_df\",\n \"rows\": 50,\n \"fields\": [\n {\n \"column\": \"query\",\n \"properties\": {\n \"dtype\": \"string\",\n \"num_unique_values\": 50,\n \"samples\": [\n \"advance directive policy define\",\n \"number of seats in rice eccles stadium\",\n \"what does the name jasmin mean\"\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"RCS\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 0.0908344955752964,\n \"min\": 0.4125649034976959,\n \"max\": 0.7750951051712036,\n \"num_unique_values\": 50,\n \"samples\": [\n 0.6688550710678101,\n 0.5449312329292297,\n 0.49656549096107483\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\": \"EAS\",\n \"properties\": {\n \"dtype\": \"number\",\n \"std\": 0.059796468065611834,\n \"min\": -0.11848333477973938,\n \"max\": 0.12401073426008224,\n \"num_unique_values\": 50,\n \"samples\": [\n -0.033763863146305084,\n -0.07555922120809555,\n -0.02830246463418007\n],\n \"semantic_type\": \"\",\n \"description\": \"\"\n }\n },\n {\n \"column\":

\"HPS\",\n   \"properties\": {\n   \"dtype\": \"number\",\n
\"std\": 7.850462293418876e-16,\n   \"min\": 0.85,\n
\"max\": 0.85,\n   \"num_unique_values\": 1,\n
\"samples\": [\n   0.85\n   ],\n   \"semantic_type\":
\"\",\n   \"description\": \"\"\n   }\n   },\n   {\n
\"column\": \"CTS\",\n   \"properties\": {\n   \"dtype\":
\"number\",\n   \"std\": 0.03234244219824685,\n   \"min\":
0.2808,\n   \"max\": 0.4152,\n   \"num_unique_values\": 47,\
n   \"samples\": [\n   0.3661\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   }\n]\n}","type":"dataframe","variable_name":"results_df"}

```
# STEP 9.4 — Quick Statistics

print("Average CTS:", round(results_df["CTS"].mean(),4))
print("Min CTS:", round(results_df["CTS"].min(),4))
print("Max CTS:", round(results_df["CTS"].max(),4))
```

```
Average CTS: 0.3446
Min CTS: 0.2808
Max CTS: 0.4152
```

```
# STEP 10 — Hallucination Injection Test
```

code - synthetic_answer = "This answer is about quantum physics and black holes." output - Average CTS: 0.343 Min CTS: 0.2783 Max CTS: 0.4111

code - synthetic_answer = "Quantum gravity explains black holes through string theory. Average CTS: 0.3446 Min CTS: 0.2808 Max CTS: 0.4152

```
# STEP 11 — Trust Calibration Analysis

# PART 1 — Grounded Evaluation Run

# ================================
# GOUNDED RUN (Correct Answer)
# ================================

grounded_results = []

for query in sample_queries:

    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score
```

```
    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    # Grounded answer = top retrieved passage
    synthetic_answer = retrieved_passages[0]

    alignment_score = evidence_alignment_score(
        synthetic_answer,
        retrieved_passages,
        model
    )

    hps = hallucination_probability(rcs, alignment_score)

    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    grounded_results.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "CTS": cts
    })

grounded_results_df = pd.DataFrame(grounded_results)

print("Grounded Avg CTS:", round(grounded_results_df["CTS"].mean(),4))
```

```
Grounded Avg CTS: 0.7788
```

```
# PART 2 — Hallucinated Evaluation Run

# ==================================
# HALLUCINATED RUN (Wrong Answer)
# ==================================

hallucinated_results = []

for query in sample_queries:

    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)
```

```
    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    # Intentional hallucination
    synthetic_answer = "Quantum gravity explains black holes through
string theory."

    alignment_score = evidence_alignment_score(
        synthetic_answer,
        retrieved_passages,
        model
    )

    hps = hallucination_probability(rcs, alignment_score)

    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    hallucinated_results.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "CTS": cts
    })

hallucinated_results_df = pd.DataFrame(hallucinated_results)

print("Hallucinated Avg CTS:",
round(hallucinated_results_df["CTS"].mean(),4))
```

```
Hallucinated Avg CTS: 0.3446
```

```
# PART 3 — Combine + Label

# Add labels
grounded_results_df["label"] = 1
hallucinated_results_df["label"] = 0

combined_df = pd.concat(
    [grounded_results_df, hallucinated_results_df],
    ignore_index=True
)
```

```python
print("Total rows:", combined_df.shape[0])
```

```
Total rows: 100
```

```python
# PART 4 — Trust Threshold Classification

# Trust threshold
threshold = 0.55

combined_df["predicted"] = (
    combined_df["CTS"] >= threshold
).astype(int)

# PART 5 — Compute Accuracy

accuracy = (
    combined_df["label"] == combined_df["predicted"]
).mean()

print("Trust Classification Accuracy:", round(accuracy,4))
```

```
Trust Classification Accuracy: 1.0
```

```python
# STEP 12 — Statistical Trust Analysis

# PART 1 — Separation Margin

grounded_mean = grounded_results_df["CTS"].mean()
hallucinated_mean = hallucinated_results_df["CTS"].mean()

separation_margin = grounded_mean - hallucinated_mean

print("Grounded Mean CTS:", round(grounded_mean,4))
print("Hallucinated Mean CTS:", round(hallucinated_mean,4))
print("Separation Margin:", round(separation_margin,4))
```

```
Grounded Mean CTS: 0.7788
Hallucinated Mean CTS: 0.3446
Separation Margin: 0.4341
```

```python
# PART 2 — Standard Deviation Overlap Check

print("Grounded STD:", round(grounded_results_df["CTS"].std(),4))
print("Hallucinated STD:",
round(hallucinated_results_df["CTS"].std(),4))
```

```
Grounded STD: 0.0541
Hallucinated STD: 0.0323
```

```python
print("Grounded Min:", grounded_results_df["CTS"].min())
print("Hallucinated Max:", hallucinated_results_df["CTS"].max())
```

```
Grounded Min: 0.6528
Hallucinated Max: 0.4152
```

```python
# PART 3 — ROC-AUC Score

from sklearn.metrics import roc_auc_score

auc = roc_auc_score(combined_df["label"], combined_df["CTS"])
print("ROC-AUC:", round(auc,4))
```

```
ROC-AUC: 1.0
```

```python
# PART 4 — Threshold Sensitivity Sweep

import numpy as np

thresholds = np.arange(0.0, 1.01, 0.05)
accuracies = []

for t in thresholds:
    preds = (combined_df["CTS"] >= t).astype(int)
    acc = (preds == combined_df["label"]).mean()
    accuracies.append(acc)

for t, a in zip(thresholds, accuracies):
    print(f"Threshold {round(t,2)} → Accuracy {round(a,3)}")
```

```
Threshold 0.0 → Accuracy 0.5
Threshold 0.05 → Accuracy 0.5
Threshold 0.1 → Accuracy 0.5
Threshold 0.15 → Accuracy 0.5
Threshold 0.2 → Accuracy 0.5
Threshold 0.25 → Accuracy 0.5
Threshold 0.3 → Accuracy 0.54
Threshold 0.35 → Accuracy 0.77
Threshold 0.4 → Accuracy 0.98
Threshold 0.45 → Accuracy 1.0
Threshold 0.5 → Accuracy 1.0
Threshold 0.55 → Accuracy 1.0
Threshold 0.6 → Accuracy 1.0
Threshold 0.65 → Accuracy 1.0
Threshold 0.7 → Accuracy 0.96
Threshold 0.75 → Accuracy 0.85
Threshold 0.8 → Accuracy 0.68
Threshold 0.85 → Accuracy 0.53
Threshold 0.9 → Accuracy 0.5
Threshold 0.95 → Accuracy 0.5
Threshold 1.0 → Accuracy 0.5
```

```python
# STEP 13 — Hard Hallucination Stress Test
```

```python
# Test Type 1 — Subtle Numerical Hallucination

# PART 1 — Subtle Wrong Answer Generator

subtle_results = []

for query in sample_queries:

    # Retrieve
    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    # Subtle hallucination (semantically similar but wrong)
    synthetic_answer = f"{query} was officially established in 1995."

    # Alignment
    alignment_score = evidence_alignment_score(
        synthetic_answer,
        retrieved_passages,
        model
    )

    # Hallucination Probability
    hps = hallucination_probability(rcs, alignment_score)

    # Composite Trust Score
    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    subtle_results.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "CTS": cts
    })

subtle_results_df = pd.DataFrame(subtle_results)
```

```python
# Print summary
print("Subtle Hallucination Avg CTS:",
round(subtle_results_df["CTS"].mean(),4))
print("Subtle Min:", round(subtle_results_df["CTS"].min(),4))
print("Subtle Max:", round(subtle_results_df["CTS"].max(),4))
```

```
Subtle Hallucination Avg CTS: 0.6714
Subtle Min: 0.5326
Subtle Max: 0.775
```

```python
print("Grounded Mean:", round(grounded_results_df["CTS"].mean(),4))
print("Hard Hallucination Mean:",
round(hallucinated_results_df["CTS"].mean(),4))
print("Subtle Hallucination Mean:",
round(subtle_results_df["CTS"].mean(),4))
```

```
Grounded Mean: 0.7788
Hard Hallucination Mean: 0.3446
Subtle Hallucination Mean: 0.6714
```

```python
# STEP 14 — Claim-Level Verification Layer

# PART 1 — Numeric Claim Extractor

import re

def extract_numbers(text):
    """
    Extract numeric tokens (years, integers, decimals)
    """
    return re.findall(r'\b\d+(?:\.\d+)?\b', text)

# PART 2 — Numeric Verification Against Evidence

def numeric_claim_verification(answer, retrieved_passages):
    """
    Check if numeric claims in answer exist in evidence.
    Returns penalty score (0 = safe, 1 = mismatch found)
    """
    answer_numbers = extract_numbers(answer)

    if len(answer_numbers) == 0:
        return 0  # No numeric claim, no penalty

    evidence_text = " ".join(retrieved_passages)

    for num in answer_numbers:
        if num not in evidence_text:
            return 1  # numeric mismatch detected

    return 0  # All numbers found in evidence
```

```python
# PART 3 — Upgrade Hallucination Probability

def hallucination_probability_v2(rcs, eas, numeric_penalty):

    gap = rcs - eas

    base_risk = 0

    if gap < 0:
        base_risk = 0.1
    elif gap < 0.05:
        base_risk = 0.3
    elif gap < 0.10:
        base_risk = 0.6
    else:
        base_risk = 0.85

    # If numeric mismatch detected, escalate risk
    if numeric_penalty == 1:
        base_risk = min(1.0, base_risk + 0.3)

    return base_risk

# PART 4 — Update Evaluation Loop (Subtle Test)

subtle_results_v2 = []

for query in sample_queries:

    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    # Subtle hallucination
    synthetic_answer = f"{query} was officially established in 1995."

    alignment_score = evidence_alignment_score(
        synthetic_answer,
        retrieved_passages,
        model
    )

    numeric_penalty = numeric_claim_verification(
```

```
        synthetic_answer,
        retrieved_passages
    )

    hps = hallucination_probability_v2(
        rcs,
        alignment_score,
        numeric_penalty
    )

    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    subtle_results_v2.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "CTS": cts
    })

subtle_results_v2_df = pd.DataFrame(subtle_results_v2)

print("Upgraded Subtle Avg CTS:",
round(subtle_results_v2_df["CTS"].mean(),4))
print("Min:", round(subtle_results_v2_df["CTS"].min(),4))
print("Max:", round(subtle_results_v2_df["CTS"].max(),4))
```

```
Upgraded Subtle Avg CTS: 0.5986
Min: 0.4725
Max: 0.7
```

```
# STEP 15 — Lightweight NLI-Based Contradiction Detection

# PART 1 — Install & Load NLI Model

!pip install transformers torch
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.10.0+cu128)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.24.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.4.1)
Requirement already satisfied: numpy>=1.17 in
```

```
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: cuda-bindings==12.9.4 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in
/usr/local/lib/python3.12/dist-packages (from torch) (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
```

```
/usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in
/usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4-
>torch) (1.3.4)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: shellingham in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: typer>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from typer-slim-
>transformers) (0.24.0)
Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)
Requirement already satisfied: certifi in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (2026.1.4)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0-
>transformers) (3.11)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: click>=8.2.1 in
```

```
/usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (0.1.2)
```

```python
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch
import torch.nn.functional as F

nli_tokenizer = AutoTokenizer.from_pretrained(
    "cross-encoder/nli-distilroberta-base"
)

nli_model = AutoModelForSequenceClassification.from_pretrained(
    "cross-encoder/nli-distilroberta-base"
)

nli_model.eval()
```


```
{"model_id":"d29c171a2c874e899b66ae4666d841b5","version_major":2,"version_minor":0}

{"model_id":"c826f29abb74478db4aca1308e521874","version_major":2,"version_minor":0}

{"model_id":"65d6fc96931448f19cd0ce2081ff88fc","version_major":2,"version_minor":0}

{"model_id":"5886ec85d19b468ebbf30dc575d20d2c","version_major":2,"version_minor":0}

{"model_id":"ff46b7a00f26469fb7cbc0c0bd0b3786","version_major":2,"version_minor":0}

{"model_id":"6a6cb100b5c147f4ad940002ee5cbb7e","version_major":2,"version_minor":0}
```

{"model_id":"9dcd7af568324219a360bf2d2d5adc34","version_major":2,"version_minor":0}

{"model_id":"994a87ede9604d63a75a28cdcb0af138","version_major":2,"version_minor":0}

```
RobertaForSequenceClassification LOAD REPORT from: cross-encoder/nli-distilroberta-base
Key                              | Status     |  |
---------------------------------+------------+--+-
roberta.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

RobertaForSequenceClassification(
  (roberta): RobertaModel(
    (embeddings): RobertaEmbeddings(
      (word_embeddings): Embedding(50265, 768, padding_idx=1)
      (token_type_embeddings): Embedding(1, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
      (position_embeddings): Embedding(514, 768, padding_idx=1)
    )
    (encoder): RobertaEncoder(
      (layer): ModuleList(
        (0-5): 6 x RobertaLayer(
          (attention): RobertaAttention(
            (self): RobertaSelfAttention(
              (query): Linear(in_features=768, out_features=768, bias=True)
              (key): Linear(in_features=768, out_features=768, bias=True)
              (value): Linear(in_features=768, out_features=768, bias=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
            (output): RobertaSelfOutput(
              (dense): Linear(in_features=768, out_features=768, bias=True)
              (LayerNorm): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
              (dropout): Dropout(p=0.1, inplace=False)
            )
          )
          (intermediate): RobertaIntermediate(
            (dense): Linear(in_features=768, out_features=3072, bias=True)
```

```
            (intermediate_act_fn): GELUActivation()
          )
          (output): RobertaOutput(
            (dense): Linear(in_features=3072, out_features=768,
bias=True)
            (LayerNorm): LayerNorm((768,), eps=1e-05,
elementwise_affine=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
      )
    )
  )
  (classifier): RobertaClassificationHead(
    (dense): Linear(in_features=768, out_features=768, bias=True)
    (dropout): Dropout(p=0.1, inplace=False)
    (out_proj): Linear(in_features=768, out_features=3, bias=True)
  )
)
```

```python
# PART 2 — NLI Scoring Function

def nli_contradiction_score(answer, retrieved_passages):
    """
    Returns contradiction probability.
    Uses top evidence passage for efficiency.
    """

    premise = retrieved_passages[0]  # strongest evidence
    hypothesis = answer

    inputs = nli_tokenizer(
        premise,
        hypothesis,
        return_tensors="pt",
        truncation=True,
        padding=True
    )

    with torch.no_grad():
        outputs = nli_model(**inputs)

    probs = F.softmax(outputs.logits, dim=1)

    # Label order: contradiction, neutral, entailment
    contradiction_prob = float(probs[0][0])

    return contradiction_prob

# PART 3 — Upgrade Hallucination Probability (v3)
```

```python
def hallucination_probability_v3(rcs, eas, numeric_penalty,
contradiction_score):

    gap = rcs - eas

    base_risk = 0

    if gap < 0:
        base_risk = 0.1
    elif gap < 0.05:
        base_risk = 0.3
    elif gap < 0.10:
        base_risk = 0.6
    else:
        base_risk = 0.85

    # Numeric mismatch escalation
    if numeric_penalty == 1:
        base_risk = min(1.0, base_risk + 0.2)

    # Contradiction escalation (weighted)
    base_risk = min(1.0, base_risk + 0.4 * contradiction_score)

    return base_risk

# PART 4 — Upgrade Subtle Test Loop

subtle_results_v3 = []

for query in sample_queries:

    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    synthetic_answer = f"{query} was officially established in 1995."

    alignment_score = evidence_alignment_score(
        synthetic_answer,
        retrieved_passages,
        model
    )
```

```
    numeric_penalty = numeric_claim_verification(
        synthetic_answer,
        retrieved_passages
    )

    contradiction_score = nli_contradiction_score(
        synthetic_answer,
        retrieved_passages
    )

    hps = hallucination_probability_v3(
        rcs,
        alignment_score,
        numeric_penalty,
        contradiction_score
    )

    cts = composite_trust_score(
        rcs,
        alignment_score,
        hps,
        stability_score
    )

    subtle_results_v3.append({
        "query": query,
        "RCS": rcs,
        "EAS": alignment_score,
        "HPS": hps,
        "Contradiction": contradiction_score,
        "CTS": cts
    })

subtle_results_v3_df = pd.DataFrame(subtle_results_v3)

print("NLI-Upgraded Subtle Avg CTS:",
      round(subtle_results_v3_df["CTS"].mean(),4))
```

```
NLI-Upgraded Subtle Avg CTS: 0.5789
```

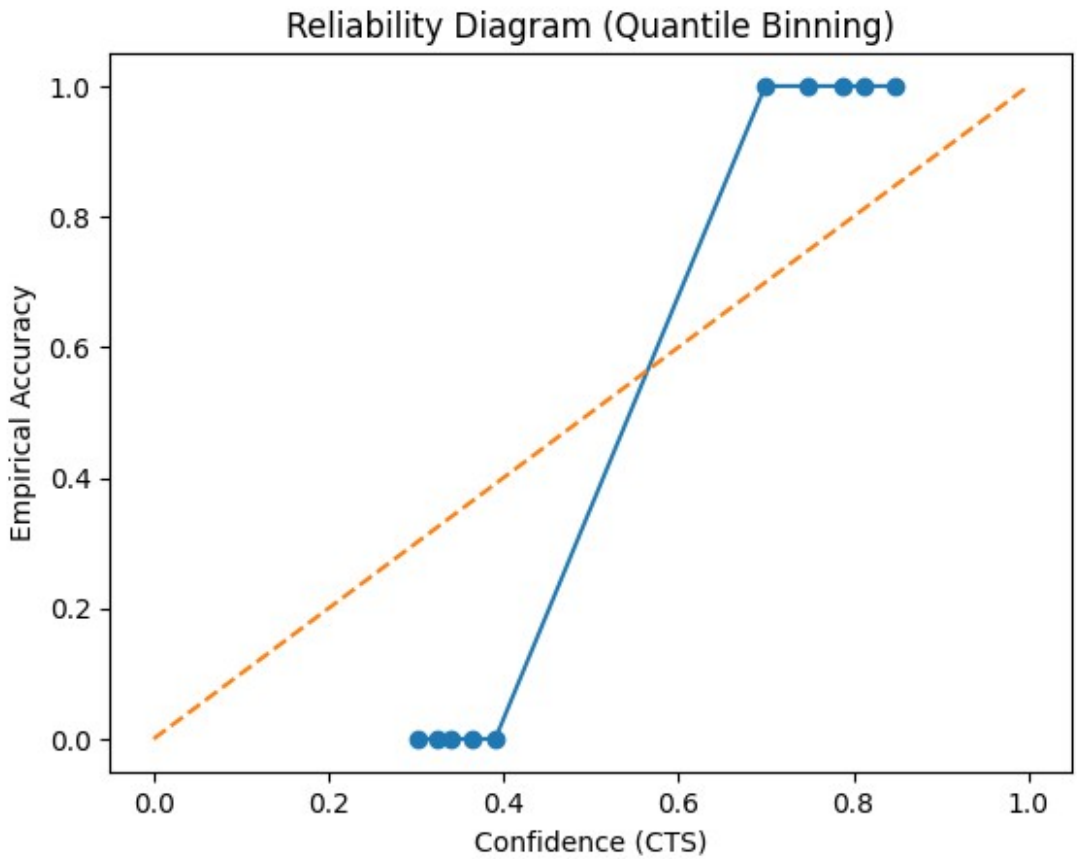
```
# STEP 16 — Calibration Curve + Expected Calibration Error (ECE)

# Step 1 — Quantile Binning

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

combined_df["prob"] = combined_df["CTS"]

# Equal-frequency binning (10 bins)
```

```python
combined_df["bin"] = pd.qcut(
    combined_df["prob"],
    q=10,
    duplicates="drop"
)

# Step 2 — Compute Calibration Stats

bin_stats = combined_df.groupby("bin").agg(
    bin_confidence=("prob", "mean"),
    bin_accuracy=("label", "mean"),
    bin_size=("label", "count")
).reset_index()

print(bin_stats)
```

```
               bin  bin_confidence  bin_accuracy  bin_size
0   (0.28, 0.319]         0.30236           0.0        10
1  (0.319, 0.332]         0.32481           0.0        10
2  (0.332, 0.354]         0.34079           0.0        10
3  (0.354, 0.371]         0.36472           0.0        10
4  (0.371, 0.534]         0.39056           0.0        10
5  (0.534, 0.727]         0.69907           1.0        10
6  (0.727, 0.768]         0.74742           1.0        10
7  (0.768, 0.796]         0.78743           1.0        10
8  (0.796, 0.833]         0.81184           1.0        10
9  (0.833, 0.878]         0.84818           1.0        10
```

```
/tmp/ipython-input-3007340560.py:1: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future
version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
  bin_stats = combined_df.groupby("bin").agg(
```

```python
# Step 3 — Plot Reliability Diagram

plt.figure()
plt.plot(
    bin_stats["bin_confidence"],
    bin_stats["bin_accuracy"],
    marker="o"
)
plt.plot([0,1],[0,1], linestyle="--")
plt.xlabel("Confidence (CTS)")
plt.ylabel("Empirical Accuracy")
plt.title("Reliability Diagram (Quantile Binning)")
plt.show()
```




```
# Step 4 — Compute ECE (Quantile Version)

ece = 0
N = len(combined_df)

for _, row in bin_stats.iterrows():
    ece += (row["bin_size"] / N) * abs(
        row["bin_accuracy"] - row["bin_confidence"]
    )

print("Quantile-Based ECE:", round(ece,4))
```

```
Quantile-Based ECE: 0.2829
```

```
# STEP 17 — Introduce Ambiguous Answers

# STEP 17.1 — Generate Ambiguous Answers

# STEP 17 — Ambiguous Evaluation (Labeled)

ambiguous_results = []

for query in sample_queries:
```

```
    # Retrieval
    query_embedding = model.encode([query])
    faiss.normalize_L2(query_embedding)

    k = 5
    scores, indices = index.search(query_embedding, k)

    rcs = float(np.mean(scores[0]))
    std_score = float(np.std(scores[0]))
    stability_score = 1 - std_score

    retrieved_passages = [df.iloc[i]["passage"] for i in indices[0]]

    base_text = retrieved_passages[0]
    words = base_text.split()

    # Type A — Partial Grounded
    truncated = " ".join(words[:len(words)//2])

    # Type B — Slight Distortion
    distorted = truncated + " It was officially recognized worldwide
in 2001."

    # Type C — Unsupported Extra Claim
    mixed = truncated + " This concept is governed by quantum gravity
principles."

    for synthetic_answer, label in [
        (truncated, 1),   # acceptable grounded
        (distorted, 0),   # factual drift
        (mixed, 0)        # unsupported addition
    ]:

        alignment_score = evidence_alignment_score(
            synthetic_answer,
            retrieved_passages,
            model
        )

        numeric_penalty = numeric_claim_verification(
            synthetic_answer,
            retrieved_passages
        )

        hps = hallucination_probability_v2(
            rcs,
            alignment_score,
            numeric_penalty
        )
```

```
        cts = composite_trust_score(
            rcs,
            alignment_score,
            hps,
            stability_score
        )

        ambiguous_results.append({
            "query": query,
            "CTS": cts,
            "label": label
        })

ambiguous_results_df = pd.DataFrame(ambiguous_results)

print("Ambiguous Avg CTS:",
round(ambiguous_results_df["CTS"].mean(),4))
print("Min:", round(ambiguous_results_df["CTS"].min(),4))
print("Max:", round(ambiguous_results_df["CTS"].max(),4))
print("Total ambiguous samples:", len(ambiguous_results_df))
```

```
Ambiguous Avg CTS: 0.7229
Min: 0.4126
Max: 0.855
Total ambiguous samples: 150
```

```
#  Combine All Classes

# Add labels if not already present
grounded_results_df["label"] = 1
hallucinated_results_df["label"] = 0

# Combine all
final_df = pd.concat(
    [
        grounded_results_df[["CTS", "label"]],
        hallucinated_results_df[["CTS", "label"]],
        ambiguous_results_df[["CTS", "label"]]
    ],
    ignore_index=True
)

print("Total final dataset size:", final_df.shape[0])
```

```
Total final dataset size: 250
```

```
# New ROC-AUC

from sklearn.metrics import roc_auc_score
```

```
auc_new = roc_auc_score(final_df["label"], final_df["CTS"])
print("New ROC-AUC:", round(auc_new,4))
```

```
New ROC-AUC: 0.8189
```

```
# Quantile Calibration Curve

import numpy as np
import matplotlib.pyplot as plt

final_df["prob"] = final_df["CTS"]

# Quantile bins
final_df["bin"] = pd.qcut(
    final_df["prob"],
    q=10,
    duplicates="drop"
)

bin_stats = final_df.groupby("bin").agg(
    bin_confidence=("prob", "mean"),
    bin_accuracy=("label", "mean"),
    bin_size=("label", "count")
).reset_index()

print(bin_stats)
```

```
               bin  bin_confidence  bin_accuracy  bin_size
0   (0.28, 0.338]        0.317724      0.000000        25
1  (0.338, 0.415]        0.371468      0.000000        25
2  (0.415, 0.661]        0.580828      0.200000        25
3  (0.661, 0.698]        0.679684      0.360000        25
4  (0.698, 0.723]        0.711380      0.240000        25
5  (0.723, 0.747]        0.734880      0.600000        25
6  (0.747, 0.776]        0.759708      0.480000        25
7  (0.776, 0.792]        0.784435      0.576923        26
8  (0.792, 0.823]        0.806071      0.708333        24
9  (0.823, 0.878]        0.839140      0.840000        25
```

```
/tmp/ipython-input-962139155.py:13: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future
version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
  bin_stats = final_df.groupby("bin").agg(
```

```
# Expected Calibration Error (ECE)

ece = 0
N = len(final_df)

for _, row in bin_stats.iterrows():
```

```
    ece += (row["bin_size"] / N) * abs(
        row["bin_accuracy"] - row["bin_confidence"]
    )

print("New Quantile-Based ECE:", round(ece,4))
```

```
New Quantile-Based ECE: 0.2586
```

```
# STEP 18 — Final Unified Evaluation

# Prepare All Labeled Data


grounded_results_df["label"] = 1
hallucinated_results_df["label"] = 0
subtle_results_v2_df["label"] = 0

# Ensure ambiguous already has label
# (If not, manually check ambiguous_results_df.columns)
print("Columns in ambiguous:", ambiguous_results_df.columns)
```

```
Columns in ambiguous: Index(['query', 'CTS', 'label'], dtype='object')
```

```
# Merge All Datasets (REAL FINAL DATASET)

# STEP 18.2 — Merge Everything

final_unified_df = pd.concat([
    grounded_results_df[["CTS", "label"]],
    hallucinated_results_df[["CTS", "label"]],
    subtle_results_v2_df[["CTS", "label"]],
    ambiguous_results_df[["CTS", "label"]]
], ignore_index=True)

print("Total unified rows:", len(final_unified_df))
print("\nLabel Distribution:")
print(final_unified_df["label"].value_counts())
```

```
Total unified rows: 300

Label Distribution:
label
0    200
1    100
Name: count, dtype: int64
```

```
# Final ROC-AUC

from sklearn.metrics import roc_auc_score

final_auc = roc_auc_score(
    final_unified_df["label"],
```

```
    final_unified_df["CTS"]
)

print("Final ROC-AUC:", round(final_auc, 4))
```

```
Final ROC-AUC: 0.8575
```

```
# Threshold Classification @ 0.60

threshold = 0.60

final_unified_df["predicted"] = (
    final_unified_df["CTS"] >= threshold
).astype(int)

accuracy = (
    final_unified_df["predicted"] ==
    final_unified_df["label"]
).mean()

print("Accuracy @ 0.60:", round(accuracy, 4))
```

```
Accuracy @ 0.60: 0.61
```

```
# Confusion Matrix

from sklearn.metrics import confusion_matrix

cm = confusion_matrix(
    final_unified_df["label"],
    final_unified_df["predicted"]
)

print("Confusion Matrix:")
print(cm)

TN, FP, FN, TP = cm.ravel()

print("\nTN:", TN)
print("FP:", FP)
print("FN:", FN)
print("TP:", TP)
```

```
Confusion Matrix:
[[ 85 115]
 [  2  98]]

TN: 85
FP: 115
FN: 2
TP: 98
```

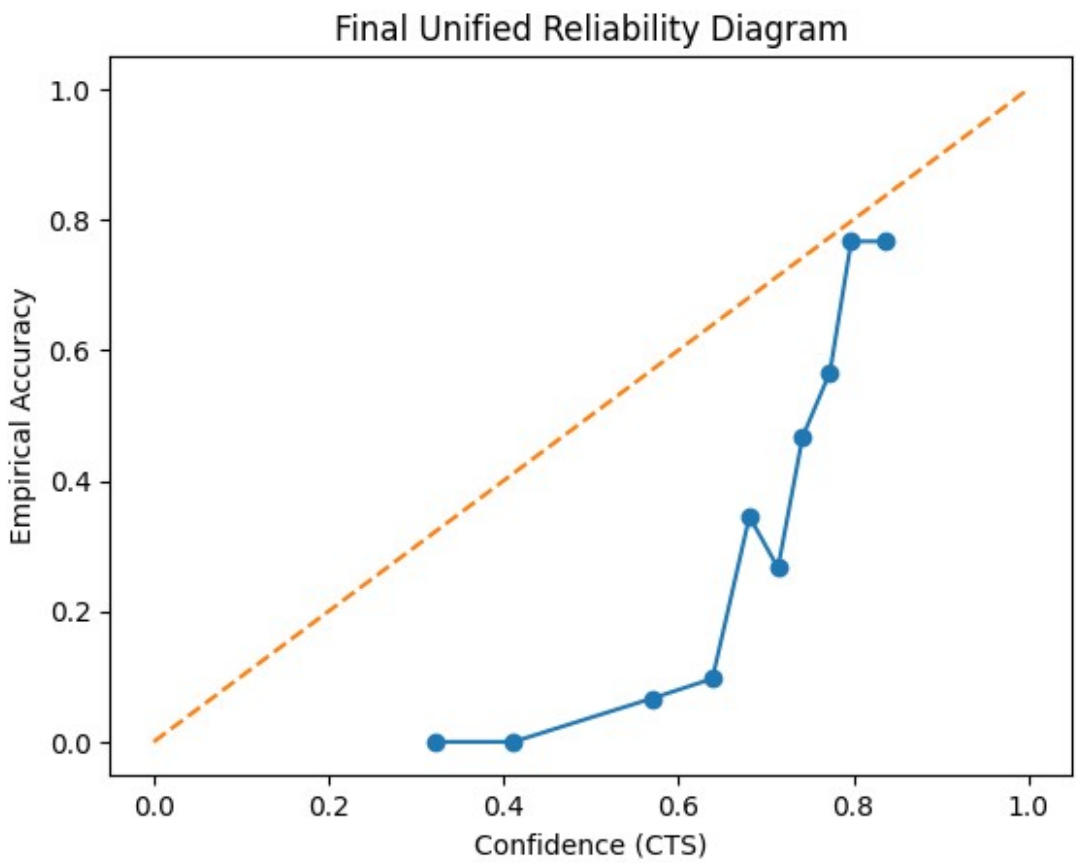
```python
# Quantile Reliability Curve

import pandas as pd
import numpy as np

final_unified_df["prob"] = final_unified_df["CTS"]

final_unified_df["bin"] = pd.qcut(
    final_unified_df["prob"],
    q=10,
    duplicates="drop"
)

bin_stats = final_unified_df.groupby("bin").agg(
    bin_confidence=("prob", "mean"),
    bin_accuracy=("label", "mean"),
    bin_size=("label", "count")
).reset_index()

print(bin_stats)
```

```
               bin  bin_confidence  bin_accuracy  bin_size
0   (0.28, 0.354]        0.322653      0.000000        30
1  (0.354, 0.532]        0.411250      0.000000        30
2  (0.532, 0.606]        0.570047      0.066667        30
3  (0.606, 0.664]        0.638800      0.096774        31
4  (0.664, 0.699]        0.681076      0.344828        29
5  (0.699, 0.727]        0.713777      0.266667        30
6  (0.727, 0.758]        0.741790      0.466667        30
7  (0.758, 0.786]        0.773237      0.566667        30
8  (0.786, 0.816]        0.797470      0.766667        30
9  (0.816, 0.878]        0.836077      0.766667        30
```

```
/tmp/ipython-input-2426912330.py:12: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future
version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
  bin_stats = final_unified_df.groupby("bin").agg(
```

```python
# Reliability Plot

import matplotlib.pyplot as plt

plt.figure()
plt.plot(
    bin_stats["bin_confidence"],
    bin_stats["bin_accuracy"],
    marker="o"
)
plt.plot([0,1],[0,1], linestyle="--")
plt.xlabel("Confidence (CTS)")
```

```
plt.ylabel("Empirical Accuracy")
plt.title("Final Unified Reliability Diagram")
plt.show()
```




```
# Expected Calibration Error (ECE)

ece = 0
N = len(final_unified_df)

for _, row in bin_stats.iterrows():
    ece += (row["bin_size"] / N) * abs(
        row["bin_accuracy"] - row["bin_confidence"]
    )

print("Final Quantile ECE:", round(ece, 4))

Final Quantile ECE: 0.3151
```